67250 IS MILIEUX · PEOPLE PROJECT

PRT Live

Rebuilding rider trust in Pittsburgh Regional Transit

Team B1 · Section B

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Executive Summary

Pittsburgh Regional Transit already collects GPS data from every bus in its fleet. The hardware works. The network exists. The software engine is capable. Despite all of this, riders cannot trust the arrival times they see in any app, official or third-party. On-time performance sits at 66 percent, well short of PRT's own 75 percent target, and roughly one in five buses can vanish entirely from tracking platforms over the course of a day.

This project investigates why a technology system that appears to function in isolation produces an experience that feels broken to the people who use it. Through primary research with PRT's Director of Transportation Technology, analysis of the BusTime API and GTFS-RT documentation, and a review of rider feedback across official and third-party platforms, we identified a sociotechnical failure pattern in which the system treats the schedule as the source of truth and GPS position as a supplementary data stream. When the schedule and reality diverge, the system defaults to silence rather than transparency.

We propose PRT Live, a three-pillar redesign that inverts this architecture. Automated trip association removes the manual driver login step that produces ghost buses. Automatic off-route detection keeps detoured buses visible on the map instead of dropping them from the feed. A transparent communication layer replaces missing data with honest confidence states, giving riders information they can act on. The solution uses infrastructure PRT already owns, which keeps it viable even under the funding constraints the agency faces through 2026.

If adopted, PRT Live would improve organizational accountability by surfacing service quality data that cannot currently be measured, restore rider trust in a public utility that many Pittsburghers depend on for daily mobility, and reframe PRT's competitive position against ride-share and personal vehicles. It would also address an equity gap, since the riders most dependent on public transit are also those most harmed by unreliable information.

Problem Identification and Context

The opportunity

Public transit is the circulatory system of a city. For the 33.3 million annual rides that Pittsburgh Regional Transit provides, the difference between a trip that works and a trip that fails often comes down to a single question: when will my bus actually arrive? PRT operates a modern intelligent transportation system, built in partnership with Clever Devices since 2000, that collects GPS position data from every bus in the fleet. And yet riders routinely describe an experience where arrival times are unreliable, buses disappear from tracking platforms without explanation, and the information displayed in apps does not match what happens on the street.

This disconnect between what the system is capable of and what riders actually see creates a compounding problem. Unreliable information drives riders toward alternatives such as ride-share services and personal vehicles. Falling ridership tightens the agency's budget. Budget pressure delays the investments that would fix the rider experience. The cycle reinforces itself, and the people most hurt by it are the riders who have the fewest alternatives.

Stakeholders

The people and organizations affected by this problem span several layers of the transit ecosystem. Each group has a different relationship to the information the system produces, and each has a different stake in seeing it improved.

Primary users

- **Daily commuters.** Riders who depend on PRT to reach work, school, and appointments. For this group, an inaccurate arrival time is not a minor inconvenience. A missed bus can mean a missed shift or a missed exam.
- **Students.** Carnegie Mellon, Pitt, and Duquesne students use PRT heavily, particularly on corridors such as the 61C, 61D, and 71A. Students make decisions in real time about whether to take the bus or walk, and those decisions rest on whether they believe the arrival estimate.
- **Transit-dependent residents.** Low-income riders, elderly riders, and people with mobility needs rely on PRT as a lifeline. For them, an unreliable tracking system is an equity issue.
- **Occasional riders.** One-off users, tourists, and event attendees who need the system to be legible even without prior familiarity.

Secondary users

- **Bus operators.** PRT drivers, represented by ATU Local 85, interact with the tracking system every time they begin a run. Their login method directly shapes what riders see.
- **Dispatchers and the disruption management team.** Staff who manually enter detours into the Clever Devices system during unplanned events such as water main breaks or street closures.

Organizational and societal stakeholders

- **PRT leadership and the Transportation Technology division.** Responsible for the tracking infrastructure and for negotiating capacity constraints with an outside vendor.
- **Clever Devices.** The 25-year sole provider of PRT's ITS hardware and software.
- **Third-party app developers.** Companies such as Transit, which consume PRT data and shape the public perception of the agency's reliability.
- **Pennsylvania state government.** Controls the Act 89 funding stream that determines PRT's operating budget.
- **The broader Pittsburgh region.** Every failed trip represents a potential rider lost to a car, with downstream effects on congestion, emissions, and economic access.

Pain points

A review of app store feedback, published reporting, and a direct interview with PRT's Director of Transportation Technology revealed a layered set of failures. Taken individually, each feels like a minor inconvenience. Taken together, they explain why a rider might check the app, see a bus coming in four minutes, and still end up waiting an hour. The following causal chain summarizes what we found:

- **Trust collapse.** Riders do not believe the arrival times they see. A 2015 Post-Gazette field test found TrueTime was accurate 96 percent of the time when GPS was functioning, but roughly 20 percent of observed arrivals never appeared on the system at all. Effective accuracy dropped to 72 percent.
- **Abandonment of the official app.** Approximately 40 percent of daily riders use Transit, a third-party app, for trip assistance. PRT's official Ready2Ride app has significantly lower adoption and has been described in app store reviews as buggy and incomplete.
- **Fare ecosystem fragmentation.** ConnectCard, Ready2Ride, and Transit operate as three parallel fare systems with no balance transfer or unified account.
- **Paid-tier inequality.** Detour information is surfaced to Transit Royale paid subscribers through a negotiated integration, but free Transit users do not see the same data even though PRT publishes it via API.

- **Silent failure.** When the system does not have data, it shows nothing. Riders cannot distinguish between a bus that is late, a bus that has been canceled, and a bus that the system simply cannot see.

Context

Several structural factors make this problem harder to solve than it appears. A 25-year vendor relationship with Clever Devices means that major software changes require contractual negotiation and vendor participation. The agency faces a 100 million dollar annual operating deficit beginning in fiscal year 2026 due to expiring Act 89 funding, with service cuts of up to 35 percent under consideration if the state legislature does not intervene. Labor dynamics add another layer, since any process change touching driver workflow must be negotiated with ATU Local 85.

These constraints shape the solution space. A proposal that requires new hardware across a 700-bus fleet is unlikely to survive a budget crisis. A proposal that depends on driver behavior change without union engagement is unlikely to succeed. The most durable solution works within the infrastructure PRT already owns, focuses on software and process rather than capital procurement, and reframes the rider experience as a public utility rather than a marketing asset to be outsourced.

Strategic Analysis

Business context

PRT is a public agency, not a profit-seeking company, which changes how we apply traditional strategy frameworks. The agency does not compete with other transit agencies in the conventional sense. It competes with the alternatives its riders consider when deciding whether to take the bus: personal cars, ride-share services such as Uber and Lyft, and the option of staying home entirely. Every rider lost to these alternatives is a competitive failure, and every ghost bus incident nudges a rider closer to that outcome.

This framing matters because it reveals where the real competitive pressure comes from. An internal metric such as on-time performance is useful, but the more important metric is rider trust. A bus that is five minutes late but clearly communicated to be five minutes late is less damaging to trust than a bus that vanishes from the tracker and reappears an hour later.

Value chain

To understand where our solution creates value, we mapped PRT's tracking system against Porter's value chain framework, adapted for a public transit agency.

Primary activities

- **Inbound logistics (data collection).** GPS pings from the IVN on each bus, automatic passenger counter data, detour inputs from the disruption management team, driver login events.
- **Operations (data processing).** Clever Devices BusTime runs the prediction engine. Each bus reports pattern-distance (pdist) and the server calculates remaining distance to each downstream stop (dstp), blending scheduled travel times with real-time speed.
- **Outbound logistics (data distribution).** GTFS-RT protobuf feeds for bus and rail, BusTime API v3 with a 10,000 request per day rate limit, CleverVision digital signage on 102 buses, SMS and QR code access at stops.
- **Marketing and sales (rider interfaces).** TrueTime web interface, Ready2Ride mobile app built by Masabi, third-party apps including Transit and Google Maps.
- **Service (rider communication).** Customer service hotline, disruption notifications, the planned Disruption Management module.

Support activities

- **Procurement.** The 25-year Clever Devices contract, Masabi for Ready2Ride, Scheidt and Bachmann for ConnectCard, GIRO for HASTUS scheduling.

- **Human resources.** Driver training on login protocols, dispatcher staffing for detour entry, the ATU Local 85 collective bargaining relationship.
- **Technology infrastructure.** The IVN hardware fleet, the cellular backhaul network, the CAD system, data centers.

Where the value chain breaks

Our research identified three specific failure points in the value chain, each of which produces downstream problems that are difficult to address in isolation.

The first failure occurs in inbound logistics. When a bus driver chooses the manual login method or bypasses the IVN entirely using only the head sign, the GPS data that flows downstream is either incomplete or unlinked to any route. Scott Vetere, PRT's Director of Transportation Technology, described this spectrum directly in our interview:

> *"It's always tracking, but it depends what it's being, what it is able to link to show to the public."*
>
> **— Scott Vetere, Director of Transportation Technology, Pittsburgh Regional Transit**

The second failure occurs in operations. The BusTime prediction engine depends on the bus's position mapping to its programmed route pattern. When the bus detours, the position no longer maps to any known geometry, and the system drops the bus rather than display data it cannot verify. Vetere confirmed that unplanned detours are one of the two biggest causes of ghost bus incidents, along with operator login behavior, and that the fix is a staffing issue more than a technology one:

> *"It would just be the manual process of putting those detours in those day-to-day incidents. So that comes down to staffing and time and all that other fun stuff."*
>
> **— Scott Vetere, on how unplanned detours reach the public tracker**

The third failure occurs in outbound logistics. PRT publishes detour data through its API, but the GTFS-RT standard has not fully adopted the detour extension, which means third-party apps on their free tiers often do not display it. Transit app users who pay for the Royale subscription see the data. Free users do not. The value chain produces information at PRT's cost, but the benefit accrues unevenly.

Where PRT Live creates value

Each pillar of our proposed solution addresses one of these failure points directly, and the value created compounds across the chain.

Automated trip association fixes inbound logistics by removing the human failure mode. When a bus leaves the depot, the system matches it to its scheduled run from HASTUS based on time, block assignment, and GPS position. The driver still logs in for payroll and fare collection purposes, but the tracking layer no longer depends on that action. This upgrade has a cascading effect. Better data in inbound logistics means better predictions in operations, a cleaner GTFS-RT feed in outbound logistics, and a more reliable experience in the marketing and sales layer.

Automatic off-route detection fixes operations by reframing how the system handles uncertainty. Rather than dropping a bus when its position diverges from the programmed route, the system flags it as off-route, continues tracking its actual position, and pauses route-based predictions until the bus returns to pattern. Dispatchers receive an automated alert to enter the detour formally, which turns a reactive process into a proactive one.

Pushing detour data into the standardized GTFS-RT Trip Modifications feed fixes outbound logistics by democratizing access. All third-party apps, including free-tier users, would receive the same detour information that paid subscribers currently see. This also has a first-party benefit, since a redesigned Ready2Ride app can use the same upgraded feed to become a credible alternative to Transit.

Competitive positioning

PRT sits in an interesting strategic position compared to its peers. Among Rust Belt transit agencies of similar size, PRT has the longest-running and most deeply integrated Clever Devices deployment. Its developer portal is strong, its GTFS-RT feeds are documented, and its BusTime API is mature. But Greater Cleveland RTA, which completed a full CAD/AVL overhaul between 2017 and 2021 with federal ATCMTD grant funding, now tracks vehicles every 15 seconds rather than PRT's 30. Cincinnati Metro, through SORTA, has recovered 117 percent of its pre-pandemic ridership by focusing on service reliability.

The comparison matters because it shows that technology upgrades translate directly into rider outcomes. PRT's infrastructure is capable. The question is whether it is being used to its full potential.

Sustainability and scalability

A solution that depends on capital procurement is not viable for PRT in the current funding environment. Our proposal is deliberately capital-light. It leverages the Clever Devices suite PRT already owns, uses HASTUS scheduling data the agency already pays to maintain, and relies on the BusTime API endpoints that already exist in the documentation. The work required is software integration, process redesign, and a negotiated update with Clever Devices. This approach is durable against budget cuts because it does not require a new line item.

The scalability argument follows the same logic. Once automated trip association and off-route detection are operational on one route, they scale across the fleet without additional hardware. The GTFS-RT improvements apply systemwide by default. Ready2Ride improvements benefit every rider who downloads the app.

Information Systems Solution: PRT Live

System concept

PRT Live is a three-layer upgrade to PRT's existing tracking stack that inverts the underlying architecture. The current system treats the schedule as the source of truth and GPS as a supplement. When GPS is available, it refines the schedule. When GPS is missing, the system falls back to scheduled data and hopes the two align. When they do not align, the rider sees nothing, or worse, sees a bus that does not exist.

PRT Live flips this relationship. GPS position becomes the source of truth. The schedule becomes the plan that reality is compared against. When GPS is missing, the system says so explicitly rather than substituting a scheduled assumption. This reframing sounds simple, but it changes what every downstream component can do.

Three pillars

Pillar 1: Automated trip association

The single biggest cause of ghost buses is the manual driver login. Vetere confirmed in our interview that PRT's current system supports three login methods, each producing a different level of data quality. An automatic login using the driver's division, payroll number, and run number links the full day's sequence of trips together and produces the cleanest data. A manual login trip-by-trip produces partial data, with gaps when the bus transitions between trips at a terminal point. A head sign entry produces no public-facing data at all.

PRT Live replaces the driver login dependency with automatic trip association. When a bus leaves the depot, the system identifies its scheduled run from HASTUS based on time of day, vehicle ID, and block assignment. GPS position confirms the route in progress. The driver's login is decoupled from tracking and used only for payroll and farebox purposes. This eliminates the head sign failure mode entirely and removes the transition gaps in manual login trips.

Implementation uses infrastructure PRT already owns. HASTUS publishes the schedule. The IVN already transmits GPS data to Clever Devices. The integration work is the association logic, which lives in software.

Pillar 2: Automatic off-route detection

When a bus leaves its programmed route, the current system drops it from the public tracker. This is the ghost bus problem in its starkest form. A rider watching the app sees a bus approaching, then sees it

vanish. The bus is still on the road, still carrying passengers, but from the rider's perspective, it has disappeared.

PRT Live adds geofence-based off-route detection. When a bus's GPS position deviates from its programmed route pattern by more than a threshold distance for more than a threshold time, the system flags the bus as off-route. Rather than dropping it, the system continues displaying its actual position on the map with a clear visual indicator, pauses its route-based arrival predictions, and triggers an automated alert to the disruption management team to confirm or enter the detour formally.

This pillar addresses what Vetere identified as the core trade-off. The disruption management module exists. The API has detour endpoints. What is missing is the automated trigger that initiates the process when a detour happens unexpectedly. Our solution is the trigger.

Pillar 3: Transparent communication layer

The final pillar is the rider-facing redesign. Every piece of data PRT collects should be communicated honestly, including uncertainty. If a bus has not transmitted its position in 45 seconds, the app should say so. If a bus is scheduled but not yet confirmed by GPS, the app should show it as unconfirmed rather than as a live estimate. If a detour is in effect, the app should announce it at the top of the stop detail screen, not bury it in an alerts menu.

This pillar also includes a redesigned Ready2Ride application. The current app is widely criticized for being slow, buggy, and fragmented from the fare system. A rebuilt version would treat rider trust as the central design goal, with confidence indicators on every arrival estimate, a clean map interface, integrated accessibility information, and native support for the new detour feed.

Pushing the standardized GTFS-RT Trip Modifications feed to third parties is part of this pillar as well. Riders who prefer Transit or Google Maps would benefit from the same improvements as Ready2Ride users, which matters because those apps are where most PRT riders live.

Key features and functionality

Across the three pillars, PRT Live introduces a set of specific features that translate architectural changes into rider experiences:

- **Confidence-tagged arrivals.** Every ETA is labeled as GPS-confirmed, last-known-position, or scheduled-unconfirmed, with distinct visual treatments on the map and stop detail screens.
- **Persistent off-route tracking.** Buses remain visible on the map during detours, with actual path shown rather than programmed route.

- **Active detour banners.** The stop detail screen surfaces any active detour affecting that stop at the top of the view, with plain-language explanation and expected resolution time.
- **Fleet-wide performance telemetry.** Because continuous GPS data is now reliable, PRT gains the ability to measure on-time performance and dwell time at a granularity that current data does not support.
- **Accessibility-aware routing.** Riders can filter arrivals for vehicles with working ramps or bike racks.
- **Subscribable alerts.** Push notifications for specific stops and routes, including automatic notice of detours affecting saved trips.

Conceptual workflow

A typical trip under PRT Live looks like this. A bus leaves the Ross Garage at 6:15 AM. Its IVN powers on with the vehicle and begins transmitting GPS. The PRT Live engine matches the bus to its scheduled run pulled from HASTUS. The bus begins its route and continuous position data flows to the prediction engine. A rider at Forbes Avenue and Morewood opens the app and sees the bus on the map, tagged as GPS-confirmed, with a 4-minute arrival estimate. Partway through the route, a water main break on Murray forces the bus onto Beacon Street. The PRT Live engine detects the deviation, flags the bus as off-route, continues tracking its actual position, and sends an alert to the disruption management team. The rider at the next stop sees the bus still moving, sees the detour banner explaining what is happening, and sees that route-based predictions are temporarily paused. A dispatcher confirms the detour in the disruption management module, which feeds the updated route back into the system, and the predictions resume with detour-adjusted estimates.

In each of these moments, the rider has information they can act on. They are not waiting for a bus that will not come. They are not staring at a blank screen wondering if the system is broken. They know what the system knows, and they know what it does not.

Feasibility, adoption, and implementation

Feasibility rests on three judgments. First, the technology exists. Clever Devices already offers a Disruption Management module, the BusTime API already includes detour endpoints, and the GTFS-RT Trip Modifications standard was adopted in 2025 with Transit app as an early integration partner. Second, the capital cost is low. The fleet hardware refresh was completed in 2020 and 2021, so the IVNs are modern. The work is software and integration, not procurement. Third, the vendor relationship is already in place, which removes the significant friction of onboarding a new CAD/AVL provider.

Adoption presents a more nuanced challenge. The Technology Acceptance Model predicts that riders will migrate to a redesigned Ready2Ride app only if they perceive it as useful and easy to use. Given that most riders currently prefer Transit, the first-party app would need to clear a high bar. Our recommendation is to focus on features that Transit cannot replicate on PRT data alone, such as native ConnectCard integration, accessibility filters, and confidence-tagged predictions sourced directly from the upgraded feed.

The labor dimension requires careful handling. Automating the login process removes a step drivers currently perform, which could be framed as surveillance reduction rather than surveillance expansion. Engagement with ATU Local 85 early in the rollout would mitigate the risk of the change being perceived as a unilateral management imposition.

Social Theory Application

Three social theories frame our analysis of how PRT Live would be used, adopted, and experienced. Each theory illuminates a different dimension of the sociotechnical problem and points to a different set of considerations in the design.

Sociotechnical systems theory

Trist and Bamforth's sociotechnical systems theory holds that organizational performance depends on the alignment between the social system and the technical system. When the two are misaligned, the whole produces worse outcomes than either would in isolation. PRT's tracking system is a textbook case. The technical system, meaning the GPS hardware, the cellular network, and the BusTime prediction engine, works correctly when evaluated on its own. The social system, meaning driver login behavior, dispatcher staffing, and rider interpretation of the output, also functions on its own terms. But the interaction between them produces failures that neither side alone would produce.

Vetere's observation that the system is always tracking but only able to show the public what it can link captures this precisely. The technical system has data. The social system determines whether that data reaches riders. When a driver logs in via a head sign rather than the automatic method, the technical system still receives GPS pings, but the social system has not created the association that lets those pings be published. The bus is tracked privately and ghosted publicly.

PRT Live is fundamentally a sociotechnical redesign. It does not add new technology so much as it removes the human dependencies that create friction between the social and technical layers. Automatic trip association makes the technical system less dependent on driver behavior. Automatic off-route detection makes the prediction engine less dependent on dispatcher availability. A transparent communication layer makes the rider experience less dependent on interpretive work that the app should be doing.

Technology Acceptance Model

Davis's Technology Acceptance Model predicts that adoption depends on perceived usefulness and perceived ease of use. The UTAUT extension, developed by Venkatesh and colleagues, adds social influence and facilitating conditions. Together these frameworks explain why Ready2Ride underperforms Transit in actual rider usage.

Perceived usefulness is low for Ready2Ride because the data it displays is unreliable. A rider who has been burned twice by ghost buses begins to distrust the app regardless of its interface quality. Perceived ease of use is also low, since the app has been criticized for slow load times, missing features, and

inconsistent integration with the fare system. Social influence points away from Ready2Ride, since the dominant peer recommendation is to use Transit. Facilitating conditions, meaning the infrastructure that supports adoption, are weak because PRT has not invested in expanding CleverVision signage or providing strong onboarding for its first-party tools.

PRT Live addresses all four factors. Perceived usefulness improves because the underlying data becomes more accurate and the uncertainty is communicated honestly. Perceived ease of use improves through the redesigned interface. Social influence can be steered through a rollout that emphasizes PRT's direct authorship of the data. Facilitating conditions improve as the GTFS-RT upgrades propagate outward to third-party apps, which means the ecosystem as a whole becomes more reliable.

The TAM lens also predicts a counterintuitive risk. A system that shows confidence states explicitly, including states of uncertainty, may initially be perceived as less useful by riders who prefer a confident number. Dietvorst's work on algorithm aversion is relevant here. People can lose trust in systems that show their limitations, even when those limitations were always present but hidden. The onboarding and the framing of the new states will matter as much as their presence.

Affordance theory

Leonardi's affordance theory, extended by Majchrzak and Markus, argues that technologies provide action possibilities that users interpret and enact in different ways. The same feature can afford different actions in different contexts. This framing is useful for thinking about what PRT Live lets riders do that the current system does not.

The current TrueTime system affords a narrow set of actions. A rider can check a predicted arrival time. A rider can see, sometimes, a bus on a map. A rider can receive an alert, if they have subscribed to one. What the current system does not afford is the ability to plan around uncertainty. A rider cannot distinguish between a bus that is genuinely late and a bus that the system has lost track of. A rider cannot see a detour in progress unless it was entered in advance. A rider cannot make a confident decision about whether to wait or walk.

PRT Live introduces new affordances by changing what the system's features mean in the rider's context. A bus on the map tagged as GPS-confirmed affords confidence in the estimate. A bus tagged as last-known-position affords informed judgment, since the rider knows the data is 45 seconds old. A bus tagged as unconfirmed affords the decision to wait or find another option. A detour banner affords a plan. In each case, the affordance emerges from the rider's interpretation of the system's honesty about its own state.

This framing also clarifies a design principle. Features do not automatically create affordances. An affordance exists only when a user interprets a feature in a way that enables action. A confidence state that is too subtle or too technical will not afford confident decision-making, no matter how accurate the underlying data is. The design must make the uncertainty legible, not just available.

Risks, trade-offs, and unintended consequences

Every design choice carries consequences beyond its primary goals. We identified five that deserve explicit attention:

- **Algorithm aversion.** Explicit confidence states may initially reduce rider trust by making uncertainty visible. The counterweight is that hidden uncertainty erodes trust over time once riders are burned, while visible uncertainty builds durable trust once riders adapt.
- **Equity and the digital divide.** A better app helps smartphone owners first. Any rollout must pair app improvements with expanded CleverVision signage, SMS enhancements, and print-schedule support so that riders without smartphones are not further disadvantaged.
- **Labor relations.** Automating the login process could be framed as surveillance if not handled carefully. Early engagement with ATU Local 85 and framing the change as the removal of a cumbersome driver task, rather than the imposition of new tracking, is essential.
- **Third-party business model impact.** Pushing detour data into the free GTFS-RT feed removes a revenue driver for Transit Royale subscriptions. This is an equity win for riders but a commercial disruption for a partner PRT depends on. The relationship should be managed rather than broken.
- **Vendor lock-in.** Deepening integration with Clever Devices extends a 25-year sole-source relationship. Over time, PRT should evaluate whether parallel integrations with analytics overlays such as Swiftly would reduce long-term dependency.

Expected Impact

Organizational impact

Internally, PRT Live changes what the agency can measure about itself. Continuous reliable GPS data across the fleet enables on-time performance analysis at a granularity that the current data cannot support. Routes with persistent dwell-time problems become visible. Drivers whose runs consistently fall behind schedule become identifiable, not for punitive reasons, but because the pattern can surface infrastructure needs such as bus lane enforcement or signal priority.

Customer service load declines as well. A significant share of rider complaints concern missing buses that were not actually missing, they were simply invisible to the tracker. Replacing silence with honest communication reduces the volume of these calls and emails.

The agency's funding position also strengthens. PRT's case for continued Act 89 support depends on demonstrating that the system delivers value to Pennsylvanians. A measurable improvement in reliability, paired with concrete rider satisfaction data, becomes an argument the agency can make to the state legislature.

Societal impact

Transit reliability is a public good with distributional effects. The riders who suffer most from ghost buses and missing detour information are the riders with the fewest alternatives. A single mother who relies on the 61C to reach a childcare pickup cannot substitute a Lyft the way a professional commuter can. When the system works reliably, the benefit flows disproportionately to those who need it most.

Environmental impact follows ridership. Every rider who chooses the bus over a car reduces vehicle miles traveled and associated emissions. An unreliable system pushes riders away from transit. A reliable system brings them back. The compounding effect over a year, across the 33 million annual rides PRT provides, is meaningful.

There is also a civic trust dimension. A well-functioning public utility reinforces confidence in public institutions more broadly. A poorly functioning one corrodes it. PRT is one of the most visible services that the Pittsburgh regional government provides, and its performance shapes how residents perceive their government's competence.

Unintended consequences

Greater transparency cuts both ways. Once PRT begins reporting accurate on-time performance data at a granular level, that data becomes a public accountability tool. This is net positive, but it also exposes the

agency to criticism during periods when performance is poor, which could happen during a major detour event, a snowstorm, or a driver shortage. The organizational response to public scrutiny of real service quality needs to be planned for.

Rider expectations may also rise faster than the system can meet them. A rider who learns to trust the new app will be unforgiving when it occasionally fails, in the way that app users are unforgiving of outages in consumer technology products. This is a manageable problem, but it requires a communication plan for the moments when something breaks.

Finally, the third-party relationship with Transit shifts. Transit has built value partly on being more reliable than PRT's own tools. If Ready2Ride becomes genuinely competitive, Transit loses some of its differentiation in the Pittsburgh market. This is not a reason to avoid the change, but the partnership terms may need renegotiation.

Team Collaboration Process

Team composition and roles

Our team of four divided responsibilities along the lines of individual strengths and interests. Patrick led the problem framing and stakeholder research, including the interview with Scott Vetere and took lead for the UI mockups. Aidan owned the strategic analysis, value chain mapping, and competitive comparison. Helen focused on the social theory application, solution design, and the website build. All team members contributed to the poster and the final document.

Roles were not rigid. We met twice a week, once to review individual progress and once to work on integration across sections. Decisions were made by consensus where possible and by lead-author authority in the individual's section where consensus could not be reached quickly.

How decisions were made

The most consequential decision of the project was reframing the problem from 'PRT needs GPS on its buses' to 'PRT has GPS but cannot use it properly.' This happened about a week into the research, after the initial hypothesis was contradicted by both published reporting and the Vetere interview. Rather than defending the original frame, we agreed to follow the evidence. The reframing strengthened the project substantially, since the revised problem statement is specific, diagnosable, and addressable by a concrete solution.

A second significant decision concerned scope. Early discussions included ambitions to redesign the fare system, integrate with micro-mobility, and propose infrastructure changes such as dedicated bus lanes. We narrowed the scope to the tracking and rider experience layer because that is where our research had the strongest evidence base and because a narrower proposal is more credible.

Conflict and resolution

One disagreement concerned the role of the Ready2Ride redesign in the solution. One view was that the first-party app should be the hero of the proposal. Another view was that pushing better data to third-party apps was a stronger leverage point because most riders already use Transit. We resolved this by keeping both in the solution but reframing them as components of a single 'transparent communication' pillar, with the recognition that the GTFS-RT feed improvements benefit every rider regardless of which app they use.

Evidence of collaboration

The supporting Google Drive folder contains our team charter, weekly meeting notes, draft iterations of each section, the full transcript and summary of the Vetere interview, and version history for the website and poster files. Together these materials document how the project evolved from an initial hypothesis to the proposal presented here.

Appendix: Research Sources and Methodology

Primary research

We conducted a semi-structured interview with Scott Vetere, Director of Transportation Technology at Pittsburgh Regional Transit, on April 17, 2026. The interview lasted approximately 20 minutes and covered the IVN system architecture, the three driver login methods and their data quality implications, the causes of ghost bus incidents, the detour handling workflow, and the relationship between PRT and third-party app developers. Direct quotes from the interview appear throughout this document where they support specific analytical claims.

Secondary research

Documentary research drew from several sources. A PublicSource investigation published in April 2024 provided foundational reporting on rider experiences with ghost buses and quotations from Vetere on the login requirement. PRT's TrueTime website and its developer portal documented the public-facing system behavior and the BusTime API specifications. PRT's FY 2024 Annual Service Report provided the 66 percent on-time performance figure and the fleet-wide ridership context. The Clever Devices product documentation and the GTFS-RT Trip Modifications specification documented the technical capabilities underlying our solution. Peer agency comparisons drew on Greater Cleveland RTA, Cincinnati Metro SORTA, and Buffalo NFTA public reporting.

Rider feedback was synthesized from app store reviews of the Pittsburgh Port Authority Bus app and Ready2Ride, spanning 2018 through 2026, which provided qualitative evidence of the pain points our solution addresses.